

Simulation and Synthesis of Fluidic Embedded Thermal Interface Materials for SiC Power Electronics

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Abstract

Silicon-carbide (SiC) MOSFET modules enable high-voltage, high-frequency power conversion in traction inverters, fast chargers, renewable-energy converters and industrial motor drives. Their switching-loss advantage can raise power density, but it also concentrates heat in a package that must survive high junction temperature, repeated power cycling and large coefficient-of-thermal-expansion (CTE) mismatch. For a packaged SiC module, the cold-side thermal stack - module baseplate or exposed substrate, thermal interface material (TIM), and liquid cold plate - is therefore a power-electronics reliability problem rather than a passive accessory.

This project evaluates a fluidic-embedded thermal interface: a sealed, electrically insulating, externally clamped interposer whose internal microchannels bring coolant closer to the module cold face. It is designed as a retrofit-capable alternative to a conventional TIM plus separate cold plate, not as a replacement for die-embedded cooling or internal module packaging. The preferred SiC implementation is a layered interface: thin compliant contact skins around a mechanically stable microchannel core, with dielectric coolant as the primary safety case and water-glycol as a comparative thermal best case.

Phase One uses conjugate heat-transfer and thermo-mechanical simulation to quantify the trade-off among interface thickness, channel geometry, pumping power, clamp load and electrical isolation. Phase Two is conditional on fabrication access and validates the most promising design with a representative SiC power-module thermal test vehicle. The report focuses on 800 V-class traction and high-voltage industrial power electronics, where compact liquid cooling and long power-cycle life are both material constraints.

Literature Survey

The SiC case combines three established but usually separate research threads: high-voltage SiC module packaging, direct liquid cooling for vehicle power electronics, and TIM durability under thermal cycling. The sources below establish the operating context and the boundary conditions for this project. Web sources were searched and scraped with Firecrawl on 9 August 2026; supplier and market sources are used for context, while the proposed design remains a research hypothesis.

Moreno, G. (2020) - Power Electronics Thermal Management (NREL / U.S. Department of Energy)

The NREL vehicle-technology project frames a 100 kW wide-bandgap inverter with an estimated 2.15 kW heat load and a high-temperature sensitivity case at 250 degC device temperature. It reports that package resistance can account for roughly 60-80% of total thermal resistance in conventional modules, and evaluates single- and double-sided dielectric-fluid cooling, jet impingement and finned surfaces.

Contribution: Provides a credible power-electronics benchmark: the project should optimize module-to-coolant resistance and pumping power, rather than report only the conductivity of a standalone material.

Review of Thermal Design of SiC Power Module for Motor Drive in Electric Vehicles (IEEE, 2024)

This SiC-specific review identifies thermal architecture as a central limitation on motor-drive size, mass, reliability and usable switching performance. It situates SiC modules in liquid-cooled automotive inverter systems, where temperature distribution and pressure loss must be considered alongside nominal thermal resistance.

Contribution: Justifies traction inverters as the primary application and supports a coupled thermal-fluidic, module-level simulation rather than a device-only heat-spreader study.

Direct Liquid Cooling of High-Performance Silicon Carbide (SiC) Power Modules (2017)

Published direct-cooling work removes the separate cold plate and TIM by placing the module in direct contact with coolant. This architecture can reduce the conventional stack resistance, but it requires package-level redesign and an integrated coolant boundary.

Contribution: Establishes the relevant benchmark while clarifying the project position: an external fluidic interface is intended for existing packaged modules where direct cooling cannot be assumed.

Bae, D. G., et al. (2017) - Embedded Two-Phase Cooling of High-Heat-Flux Electronics on SiC Using a FEEDS Manifold-Microchannel Cooler

Bae and colleagues reported embedded two-phase cooling on SiC using a manifold-microchannel delivery structure. The work demonstrates the high heat-extraction potential of near-junction microfluidics, while also showing that this class of solution relies on specialized packaging and fluid delivery rather than a retrofit interface.

Contribution: Motivates channel/manifold optimization, but is intentionally outside the proposed product boundary because it changes the semiconductor-package architecture.

Thermal Management of Wide-Bandgap Power Semiconductors: Strategies and Challenges in SiC and GaN Power Devices (Electronics, 2025)

The review covers direct liquid cooling, jet impingement, phase change and embedded microchannels for wide-bandgap devices. Its central design implication is that thermal-interface and package resistances remain consequential even when the semiconductor has favorable bulk thermal properties.

Contribution: Supports treating the interface as part of the full junction-to-coolant path and comparing against practical cold-plate architectures.

Wolfspeed - SiC Power Module Reliability: Power Cycling and Lifetime Modeling Approach (accessed 2026)

The module-reliability discussion distinguishes short power cycles that emphasize die-top and die-attach interfaces from longer cycles that heat the substrate, solder joints and baseplate attachments. It identifies power-cycle variables such as junction-temperature swing, peak junction temperature and pulse duration, and uses a 20% increase in thermal resistance as one thermal failure criterion.

Contribution: Defines the transient reliability cases that a fluidic interface must survive; a low steady-state resistance alone is insufficient evidence of value.

Degradation of Thermal Interface Materials for High-Temperature Power Electronics Applications (Microelectronics Reliability, 2013)

This open-access study examined TIM aging in high-temperature half-bridge power-switch conditions. After isothermal aging at 170 degC for 90 days, the reported specific thermal resistance of the tested TIM configurations was not substantially different from a no-TIM contact pair in the stated samples, illustrating that interface behavior and degradation must be measured, not inferred from datasheet conductivity.

Contribution: Makes thermal aging, pressure retention and resistance drift mandatory Phase One and Phase Two metrics.

Wang, Y. (2024) - Thermal Material Trends Driven by SiC Adoption in EV Power Electronics (IDTechEx)

The industry review places SiC MOSFET adoption in the move toward 800 V-and-above EV architectures. It notes that TIMs are commonly placed between the baseplate and heatsink, and reports a 2024 typical TIM conductivity range of 2.5-3 W/mK for this role, with a market push toward higher-conductivity materials and altered cooling architectures.

Contribution: Confirms that the proposed interface targets a real external TIM location, while its performance must be assessed at the system level rather than against conductivity alone.

Literature Gap

The reviewed literature has mature solutions at two ends of the stack: package-integrated direct cooling that eliminates the TIM but requires a redesigned module, and improved TIMs that retain a separate cold plate. It does not establish, for off-the-shelf SiC power modules, the combined thermal-fluidic, mechanical and insulation design map for a sealed microchannel interface placed at the cold-side TIM location. In particular, there is limited evidence that a single compliant material can also provide robust coolant sealing and high-voltage isolation over automotive-style power cycling.

The narrow research question is therefore: **under realistic SiC module clamp loads and cycling, can a layered, electrically insulated fluidic interface reduce module-to-coolant thermal resistance enough to offset its added thickness, flow resistance and reliability risk?** This is a technical hypothesis, not a claim of patent freedom or commercial novelty. A formal prior-art and freedom-to-operate search is a prerequisite before commercialization.

Objectives

- Define a SiC power-module use case centered on a 1200 V-class traction-inverter half bridge, with industrial drives and fast chargers as secondary application cases.
- Build a conjugate heat-transfer model for an externally clamped, fluidic-embedded TIM and benchmark it against a conventional TIM plus liquid cold plate of equivalent footprint.
- Sweep contact-skin thickness, microchannel width, pitch, core thickness, coolant flow rate and manifold layout to create a module-to-coolant resistance versus pressure-drop design map.
- Evaluate thermo-mechanical compliance, clamp-pressure sensitivity, electrical isolation, seal stress and resistance drift under short- and long-pulse SiC power-cycling conditions.
- Identify whether a layered interface is necessary: compliant ceramic-filled contact skins for conformity, plus a sealed structural microchannel core for flow control and isolation.
- Select a geometry for conditional fabrication and testing using a representative thermal test vehicle or a selected commercial SiC module with no internal package modification.

SiC Power-Electronics Scope

Design item	Project decision	Reason
Primary application	800 V vehicle traction inverter using a 1200 V-class SiC module	High power density, liquid cooling and repetitive drive-cycle loading make the cold-side interface consequential.
Secondary cases	DC fast charger, renewable-energy converter and industrial motor drive	All use packaged SiC power modules, but their coolant temperature and duty cycle shall be separate parameter sets.
Target location	External module cold face / baseplate-to-coolant interface (TIM2 location)	Preserves the purchased module and avoids claims that require die or substrate redesign.
Coolant	Single-phase dielectric coolant is the primary safety case; water-glycol is a comparative best-case model	A conductive coolant can improve heat transfer but increases the consequence of a channel or seal failure near high-voltage hardware.
Temperature cases	Supplier-rated normal operation plus high-temperature sensitivity; power-cycle cases separated into short and long pulses	Avoids presenting the DOE 250 degC high-temperature target as the continuous rating of every commercial SiC module.

Reference Architecture and Simulation Plan

The baseline stack is: SiC power module - conventional TIM - metallic liquid cold plate. The proposed stack is: SiC power module - thin compliant dielectric contact skin - sealed microchannel core - thin compliant dielectric contact skin - external manifold. The model will include the real module footprint and heat map of the selected reference device; it will not substitute a uniform heat flux unless that simplification is reported and separately checked.

Parameter family	Initial simulation range	Reported output
Contact layer	50-250 um per skin; ceramic-filled compliant dielectric candidates	Contact resistance, clamp-load sensitivity, interface stress
Channel core	0.5-1.5 mm overall core; 0.2-1.0 mm hydraulic channel scale	Conduction penalty, local heat-transfer coefficient, seal stress
Layout	Pitch, channel count, inlet/outlet position, manifold taper	Peak temperature, temperature non-uniformity, pressure drop
Coolant	Dielectric fluid primary; water-glycol comparison; flow-rate sweep	Pumping power, outlet temperature rise, fail-safe sensitivity
Cycling	Short and long power pulses; representative delta-Tj and peak-Tj cases from the selected module	Thermal-resistance drift, warpage, damage-prone locations

The first model should be a coupled, single-phase conjugate heat-transfer study in COMSOL or OpenFOAM. A compact thermal model of the selected SiC module can supply the junction-to-case portion, while the CFD model resolves the external interface. Thermo-mechanical analysis then evaluates clamp load and CTE mismatch using the temperature field. Two-phase cooling is explicitly deferred: it may improve heat transfer but adds flow-instability, orientation, sealing and validation risk before the value of the single-phase interface is known.

Success and Stop Criteria

- Thermal: lower total module-to-coolant resistance and lower peak temperature than the conventional TIM-plus-cold-plate baseline at equal footprint and defined pumping-power limit.
- Hydraulic: pressure drop and pumping power compatible with the selected inverter or converter cooling loop; report both, not one alone.

- Reliability: no unacceptable resistance drift, seal stress, warpage or loss of contact pressure after representative thermal-cycle simulations and subsequent test cycles.
- Electrical safety: retain required isolation resistance and dielectric withstand capability after thermal exposure and leak/pressure testing; actual limits come from the selected module and system safety requirements.
- Stop criterion: do not prototype a geometry whose cooling benefit exists only with impractical pumping power, non-representative contact pressure, a conductive-coolant failure risk, or degraded insulation margin.

Timeframe

Phase	Work and expected status
Phase One	Complete the SiC-module literature review, select one 1200 V-class reference module or validated thermal test vehicle, and document its permitted temperature, mounting and coolant boundary conditions. Establish the conventional TIM-plus-cold-plate baseline.
Phase Two	Run conjugate heat-transfer sweeps across contact thickness, channel geometry, coolant and flow rate. Produce a resistance-versus-pressure-drop map and identify configurations that remain feasible under a clamped external interface.
Phase Three	Add thermomechanical and electrical-isolation checks. Evaluate clamp-pressure sensitivity, CTE-induced stress, seal margins and short-/long-pulse thermal cycling. Document negative results as design limits.
Phase Four	Conditional: explore fabrication through CeNSE/INUP or another suitable prototyping route. Fabricate only the geometry that meets the Phase Two and Three gates; conduct pressure, leak, thermal-resistance and isolation tests.
Phase Five	Compile the final simulation and test report. Position any future funding proposal around SiC power-module thermal reliability and compact cooling, with a separate prior-art and freedom-to-operate assessment before commercialization.

Project Budget

No dedicated budget is required for Phase One if existing CFD and structural-simulation access is available. Phase Four remains unbudgeted until a fabrication route, coolant choice, test vehicle and required isolation standard are confirmed. A prototype budget must include materials, manifold fittings, pressure/leak testing, thermal characterization and electrical-isolation testing; cleanroom access is not assumed as a prerequisite for an initial externally clamped prototype.

References

- 1) Moreno, G. (2020). *Power Electronics Thermal Management*. National Renewable Energy Laboratory / U.S. Department of Energy. [\[source\]](#)
- 2) *Review of Thermal Design of SiC Power Module for Motor Drive in Electric Vehicles*. IEEE, 2024. [\[source\]](#)
- 3) *Direct Liquid Cooling of High-Performance Silicon Carbide (SiC) Power Modules*. 2017 publication record. [\[source\]](#)
- 4) Bae, D. G., Mandel, R. K., Dessiatoun, S. V., Rajgopal, S., Roberts, S. P., Mehregany, M. and Ohadi, M. M. (2017). *Embedded two-phase cooling of high heat flux electronics on silicon carbide (SiC) using a FEEDS manifold-microchannel cooler*. ITherm. [\[source\]](#)
- 5) *Thermal Management of Wide-Bandgap Power Semiconductors: Strategies and Challenges in SiC and GaN Power Devices*. Electronics, 2025. [\[source\]](#)
- 6) Wolfsped. *SiC Power Module Reliability: Power Cycling and Lifetime Modeling Approach*. Accessed 9 August 2026. [\[source\]](#)
- 7) *Degradation of thermal interface materials for high-temperature power electronics applications*. Microelectronics Reliability 53(12), 1933-1942 (2013). [\[source\]](#)
- 8) Wang, Y. (2024). *Thermal Material Trends Driven by SiC Adoption in EV Power Electronics*. IDTechEx. [\[source\]](#)
- 9) Indian Nanoelectronics Users' Programme (INUP), Centre for Nano Science and Engineering, IISc Bengaluru. [\[source\]](#)